MSEP Area 2025 Data Summaries

Mississippi Sound Estuary Program

October 30, 2026

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Introduction

This website is a collection of documents summarizing the condition of the Mississippi Sound and its watershed in 2025, as compared to "baseline" conditions (which we also need to define, at appropriate spatial scales for each parameter). This will be the technical supporting information for what ultimately goes into a State of the Sound report. The State of the Sound will be public-facing, whereas the intended audience for this document is largely technical.

The outputs are compiled as a 'quarto book'. Each chapter is produced by a separate `.qmd` file.

The purpose of the book format is to make the outputs readily findable and accessible. It also serves as an outline for future undertakings like a complete baseline characterization of these parameters and future State of the Sound reports.

You can download a pdf version of this entire book by clicking on the pdf icon on the right side of the toolbar at the top of this page.

Part I

Weather and Climate-related variables

1 Precipitation

1.1 Overall Summary

2025 was normal-to-dry across the MSEP watershed, with annual total precipitation within 5% of the 30-year average in 4 of the 14 subbasins, and 10-18% lower than normal in the other 8 subbasins. See Figure 1.5 and Figure 1.3 in particular.

1.2 Supporting Information

1.2.1 Annual Totals, Averaged to HUC8

Annual totals were calculated from monthly precipitation data obtained from the PRISM Group at Oregon State University, https://prism.oregonstate.edu which were accessed 29 July 2026.

1.2.1.1 Map

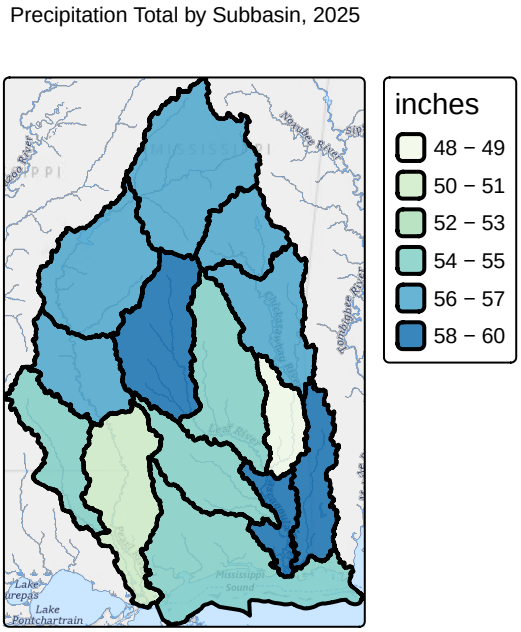


Figure 1.1: Map of annual precipitation total (inches), averaged to the HUC-8 (subbasin) level.

1.2.1.2 Time Series Graph

How does 2025 compare to past years? This graph uses data compiled in the `msepBaselineData` package.

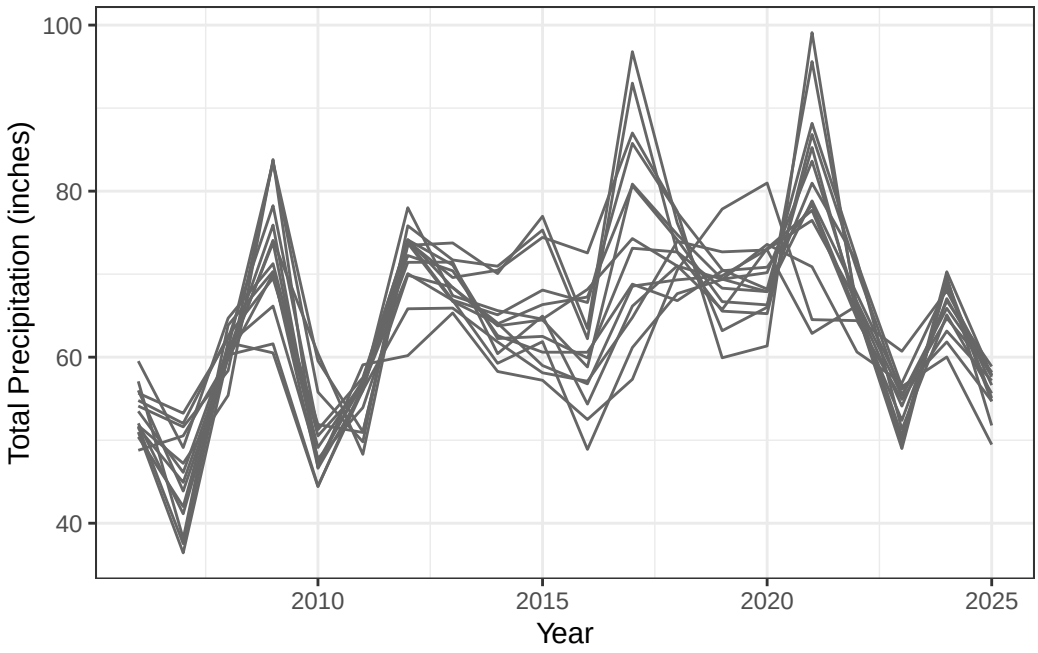


Figure 1.2: Time series of annual precipitation total (inches) by HUC-8 (subbasin). Each line represents a subbasin.

1.2.2 Annual Anomalies

Because the climate normal on the coast is upwards of 60 inches, and it looks like the coast was in the mid-50s range in 2025, which is a little drier than normal. Climate normals in the northern part of the watershed are in the mid-to-upper 50s so they look about typical for 2025. I'm eyeballing here and the color scale is slightly different, but see the MSEP Characterization chapter on precipitation for the details I have compiled in the past (climate normals from NCEI).

1.2.2.1 Map

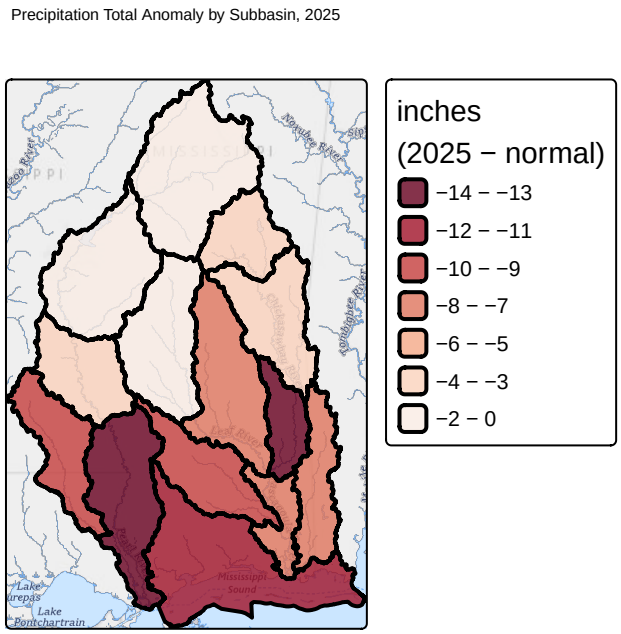


Figure 1.3: Map of 2025 differences from 30-year average precipitation, in inches, averaged to the HUC-8 (subbasin) level. Darker colors indicate totals further from the long-term average.

1.2.2.2 Time Series Graph

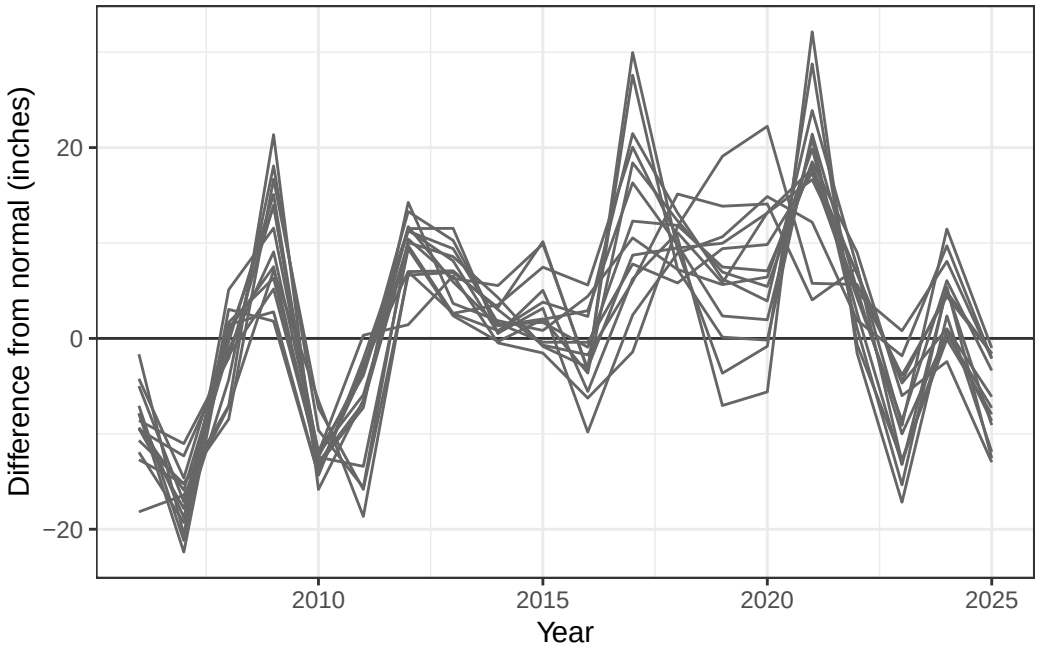


Figure 1.4: Time series of annual anomalies (difference between that year and the 30-year climate normal, in inches) by HUC-8 (subbasin). Each line represents a subbasin.

The y-axis may not be that meaningful if you don't already know that the annual normal amount is around 60 inches. So let's look at the difference as a percent.

From this 20-year time series, we can see that generally things tend to be similar (compared to normal) across all the HUC8s in the watershed. Some years have more variability, like 2019 and 2020. We had a fairly long stretch from 2012-2021 or so where it generally was normal to wetter-than-normal. 2025 was generally drier than normal across the watershed; drier in some HUC8s than others. As seen from the map, the southern counties were the ones that generally were much drier than normal.

1.2.2.3 Interactive time series

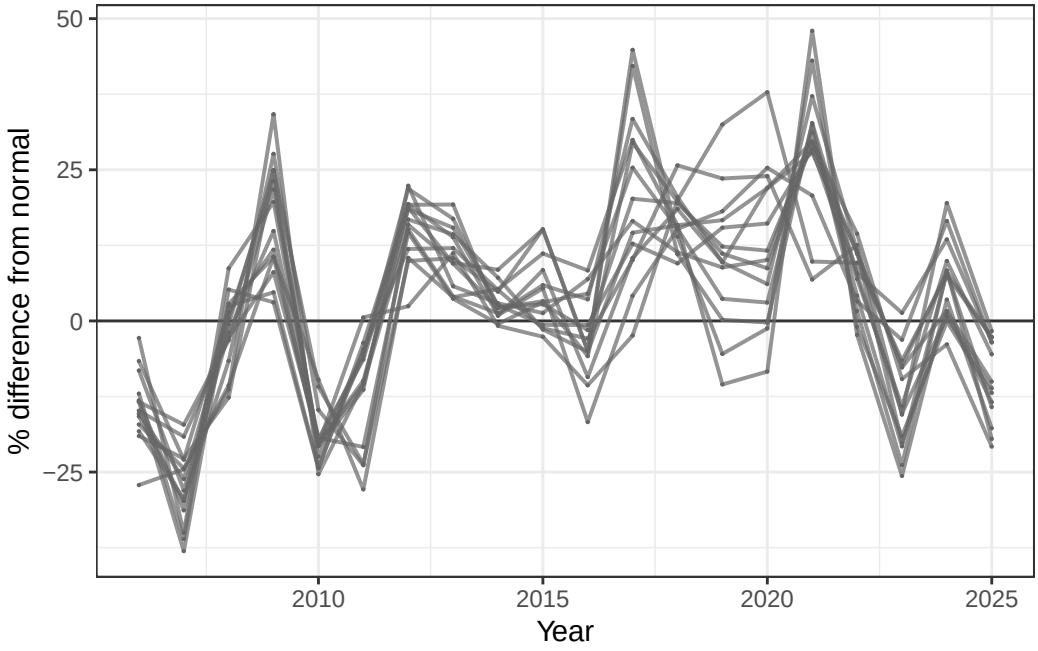


Figure 1.5: Time series of differences from 30-year average precipitation, by HUC-8 (subbasin). Each line represents a subbasin. In the HTML version of this document, hovering over a line shows which subbasin it represents, and hovering over a point shows the precipitation total and anomaly for that year in the subbasin.

1.2.2.4 Boxplot time series

Each box represents a single year; overlaid points represent subbasins.

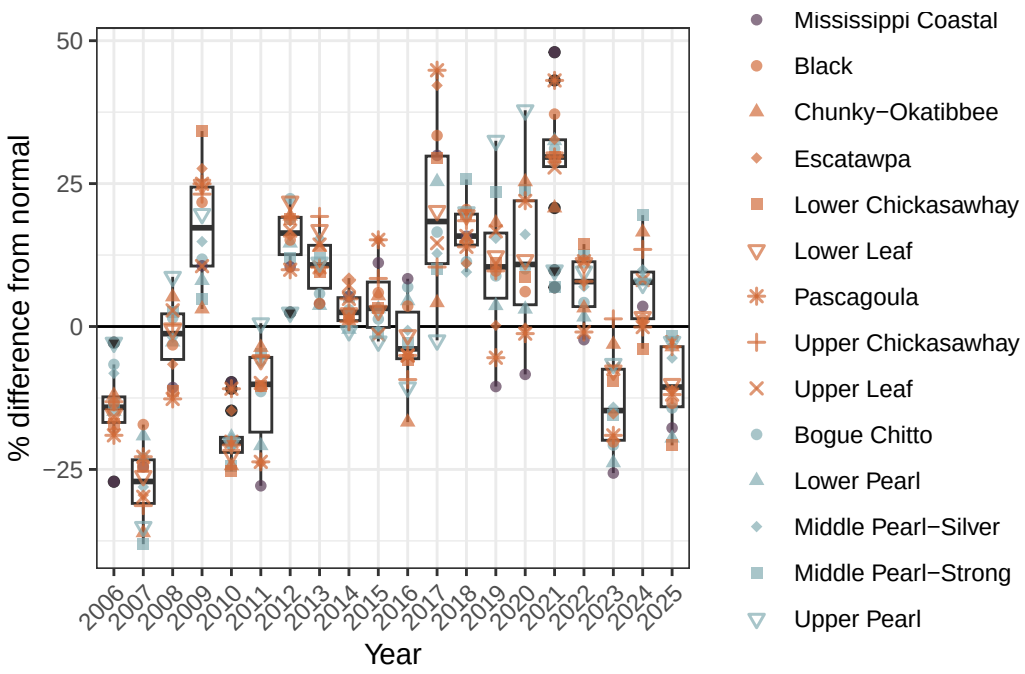


Figure 1.6: Boxplot representing the distribution of rainfall totals by subbasin by year. Each box represents a year; each overlaid point represents a subbasin. Each color represents a basin, with shapes within colors representing the subbasins. Whiskers extend to the quantiles +/- 1.5 times the IQR.

Here is the same plot, but all points are the same color so it's simpler.

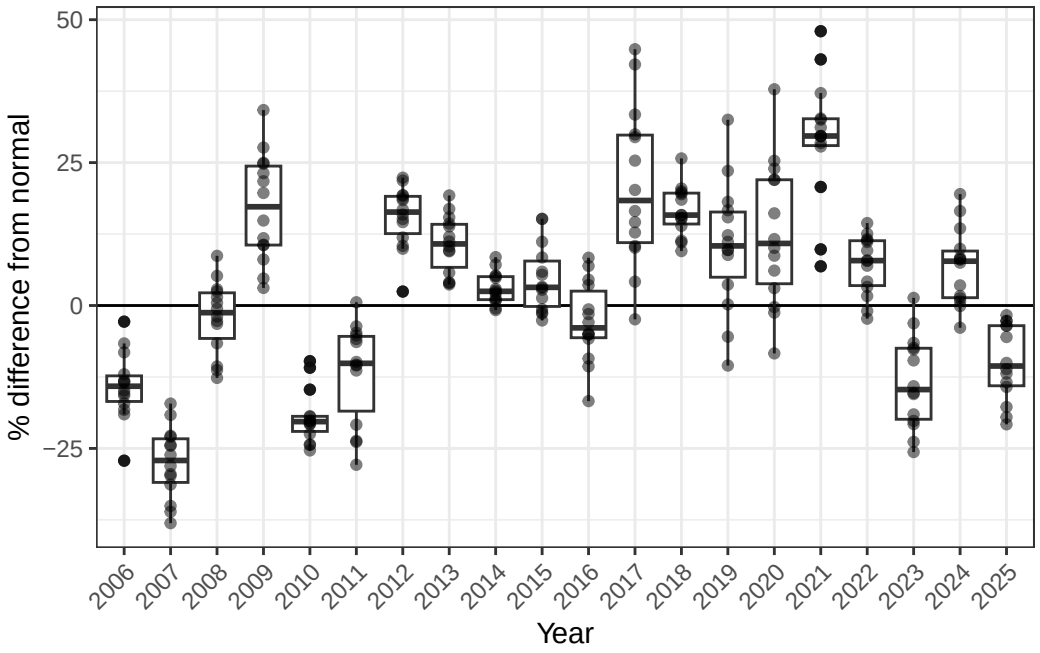


Figure 1.7: Boxplot representing the distribution of rainfall totals by subbasin by year. Each box represents a year; each overlaid point represents a subbasin. Whiskers extend to the quantiles +/- 1.5 times the IQR.

1.2.3 Monthly Totals

Monthly data are from the PRISM Group at Oregon State University, https://prism.oregonstate.edu and were accessed 29 July 2026. They are again averaged to HUC8.

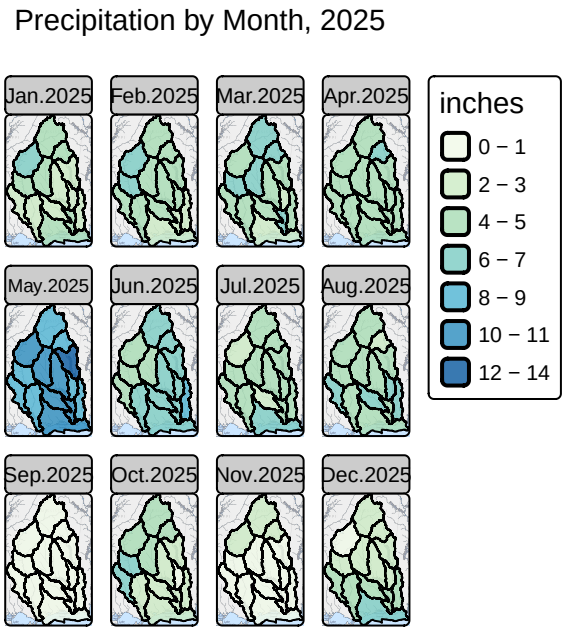


Figure 1.8: Map of monthly precipitation totals (inches), averaged to the HUC-8 (subbasin) level.

May seems to have been the wettest month in 2025, and that's not really typical - actually in the climate normals (see them here), the color scale only goes up to 8.5 inches per month, and May 2025 exceeded that.

1.2.4 Monthly Anomalies

From long-term climate normals, as provided by the PRISM Climate Group at Oregon State University and processed into the `msepBaselineData` package's `precip_normals_huc8avg` dataset.

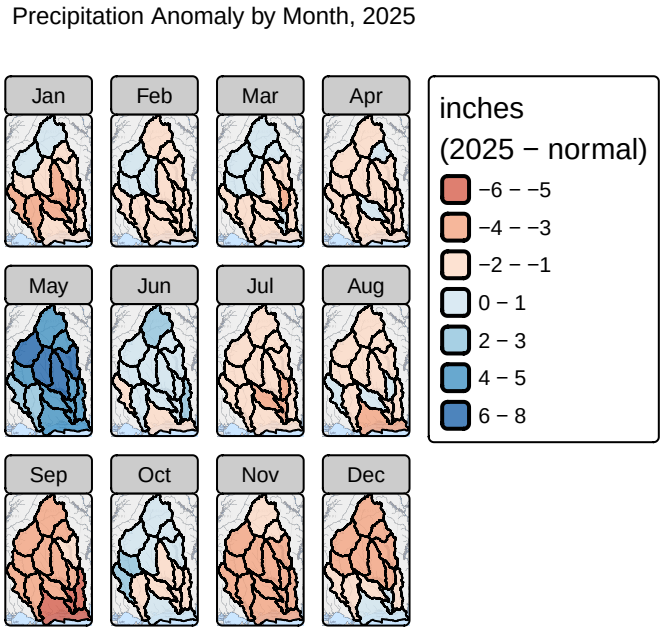


Figure 1.9: Map of monthly precipitation anomalies (difference from 30-year climate normals, in inches), averaged to the HUC-8 (subbasin) level.

1.2.5 Table of Anomalies

Table 1.1: Precipitation Anomalies by Subbasin, 2025 (inches)

Basin	HUC8	Subbasin	Annual	Jan	Feb	Mar	Apr	May	Jun	
Coastal Streams	Mississippi Coastal	03170009	-11.85	-1.32	-0.72	-1.56	-1.52	4.72	-0.35	-1
Pascagoula River	Chunky-Okatibbee	03170001	-2.18	-0.19	-1.35	1.14	0.18	5.09	1.72	-1
Pascagoula River	Upper Chickasawhay	03170002	-2.24	-1.88	-0.30	-0.01	-0.31	7.88	1.42	-0
Pascagoula River	Lower Chickasawhay	03170003	-13.00	-2.96	-0.58	-2.36	-0.86	4.85	0.41	-2
Pascagoula River	Upper Leaf	03170004	-1.60	-1.38	0.20	0.29	-0.62	7.73	1.01	-0
Pascagoula River	Lower Leaf	03170005	-6.05	-2.50	-0.35	-1.33	-0.04	6.71	0.86	-0
Pascagoula River	Pascagoula	03170006	-7.92	-1.22	-0.68	0.55	-0.76	5.22	0.94	-2
Pascagoula River	Black	03170007	-8.61	-2.52	-0.32	-1.53	0.15	5.77	1.05	-2
Pascagoula River	Escatawpa	03170008	-7.32	-1.95	-0.85	-0.63	-0.04	4.82	2.00	-1

Pearl River	Upper Pearl	03180001	-1.56	0.31	-1.12	0.69	-0.54	4.40	2.57	-1
Pearl River	Middle Pearl-Strong	03180002	-0.95	0.50	0.48	0.16	-0.84	6.51	1.33	-1
Pearl River	Middle Pearl-Silver	03180003	-3.29	-1.84	0.30	0.57	-0.43	4.86	0.34	-1
Pearl River	Lower Pearl	03180004	-12.69	-2.61	-0.72	-1.38	-0.52	3.66	0.08	-1
Pearl River	Bogue Chitto	03180005	-8.94	-2.82	-1.11	-0.30	-0.29	3.38	-0.90	-1

1.2.6 Tropical Events

1.2.7 Drought Index

1.2.8 El Nino

About

The `.qmd` file that generated this section was: **precipitation.qmd**.

For information on data sources and processing, see *appendix cross-reference here*.

2 Freshwater Inflows

2.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "River inputs were about average this year", or "Pearl River inputs were higher than normal, while Pascagoula River inputs were lower than normal", or "Flows from both main rivers were higher than normal this year, and the Bonnet Carre spillway was open for a substantial amount of time, causing XYZ impacts".

2.2 Supporting Information

2.2.1 Rivers

Pearl and Pascagoula River USGS gauges

2.2.2 Mobile Bay

?? certainly affects the Eastern Sound, even if exact amount is in question (estimates vary; I have some references if we want to talk about this) so probably worth noting.

2.2.3 Bonnet Carre

- number of openings
- number of days open
- Volume released

About

The `.qmd` file that generated this section was: **freshwater_inflows.qmd**.

For information on data sources and processing, see *appendix cross-reference here*.

Part II

People and Land Use

3 Population

3.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Population has been steady in the watershed for the last 10 years"

3.2 Supporting Information

See population calculations in the msep_geospatialCharacterization repository, and repeat them for current year vs. 5, 10, 20 years ago.

About

The `.qmd` file that generated this section was: **population.qmd**.

For information on data sources and processing, see *appendix cross-reference here*.

4 Land Use / Land Cover

4.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "The coastal counties of the watershed have become more developed in the last 10 years, while developed landcover in the more inland counties has remained relatively stable"

4.2 Supporting Information

About

The `.qmd` file that generated this section was: **landuse_landcover.qmd**.

For information on data sources and processing, see *appendix cross-reference here*.

Part III

Water Conditions

5 Salinity

5.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Salinity was lower than normal, and this was more pronounced in the Eastern Sound than the rest of the area." Tie back to Chapter 1 and Chapter 2 chapters.

5.2 Supporting Information

5.2.1 Deployed Sensors

USGS/DMR hydrological stations, DMR sensor network, USM-GCRL stations (in the future), Grand Bay NERR SWMP stations

5.2.2 DMR fisheries and shrimp sampling

5.2.3 Mississippi Coastal Assessment

5.2.4 Other salinity data in the area

About

The `.qmd` file that generated this section was: **salinity.qmd**.

For information on data sources and processing, see *appendix cross-reference here.*

6 Dissolved Oxygen

6.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Hypoxia was more (or less) prevalent than usual in the eastern Sound and its bays, and about normal elsewhere along the coast. In freshwater streams and rivers, hypoxia was about normal."

6.2 Supporting Information

6.2.1 DEQ data

Inland sampling network programs

6.2.2 Deployed Sensors

USGS/DMR hydrological stations, DMR sensor network, USM-GCRL stations (in the future), Grand Bay NERR SWMP stations

6.2.3 DMR fisheries and shrimp sampling

6.2.4 Mississippi Coastal Assessment

6.2.5 Other salinity data in the area

About

The `.qmd` file that generated this section was: **dissolved_oxygen.qmd**.

For information on data sources and processing, see *appendix cross-reference here*.

7 Turbidity / Suspended Solids

7.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Turbidity was lower than normal (clearer water!) in the rivers and streams, while Total Suspended Solids were higher than normal across the Sound."

7.2 Supporting Information

7.2.1 Turbidity - Deployed Sensors

DMR sensor network, USM-GCRL stations (in the future), Grand Bay NERR SWMP stations

7.2.2 TSS - DEQ Beach Sampling

7.2.3 TSS - Mississippi Coastal Assessment

7.2.4 Other data in the area

Grand Bay SWMP has been doing TSS since ~2021

About

The `.qmd` file that generated this section was: **turb_tss.qmd**.

For information on data sources and processing, see *appendix cross-reference here.*

8 Nutrients

May want to split this into N and P; there are different species of each and that might be a lot for one chapter.

8.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Nitrogen and phosphorus concentrations were higher than normal, and this was more pronounced in the Eastern Sound than the rest of the area. This coincides with extreme Bonnet Carre inputs."

8.2 Supporting Information

8.2.1 DEQ Beach Sampling

8.2.2 DEQ Freshwater Programs

8.2.3 Mississippi Coastal Assessment

8.2.4 Other data in the area

Grand Bay SWMP

About

The `.qmd` file that generated this section was: **nutrients.qmd**.

For information on data sources and processing, see *appendix cross-reference here.*

9 Chlorophyll *a*

Make sure to note whether programs are using "chl corrected for phaeophytin" or "not corrected for phaeophytin".

9.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Chlorophyll was about normal across coastal waterbodies this year."

9.2 Supporting Information

9.2.1 DEQ Beach Sampling

9.2.2 DEQ Freshwater Programs

9.2.3 Mississippi Coastal Assessment

uses uncorrected chl

9.2.4 Other data in the area

Grand Bay SWMP; uses corrected chl

About

The `.qmd` file that generated this section was: **chla.qmd**.

For information on data sources and processing, see *appendix cross-reference here.*

Part IV

Pollutants/Contaminants

10 Beach Advisories

This section will be based on data from beaches.mdeq.ms.gov.

10.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Overall, the number of beach advisory days was lower than normal, and at *example hot spot stations* in particular, the number has been decreasing."

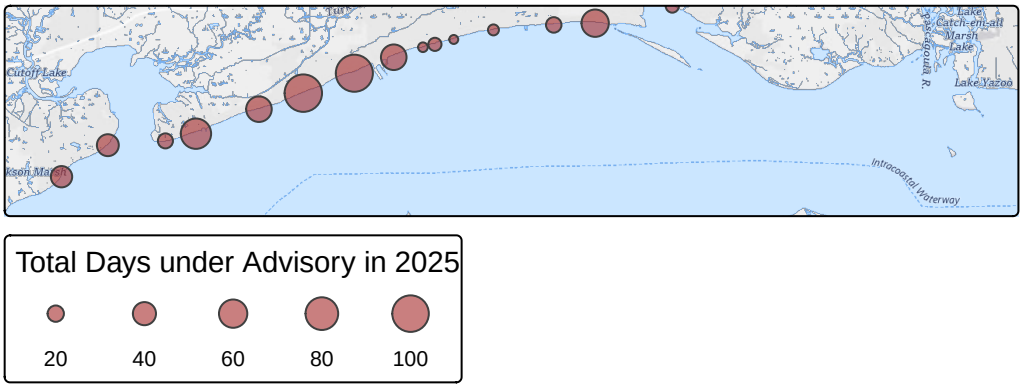
10.2 Supporting Information

10.2.1 DEQ Beach Sampling

Make a map where each station is represented by a point, and the point size is based on the number of days that station was under advisory in the year.

no. of beach closure days by station (see PPBEP's online map)

- Can download advisory/closure info from DEQ: https://beaches.mdeq.ms.gov/Historical/HCA
- Per station, calculate # advisories/closures, # days under advisory/closure (total), mean or median # days per advisory/closure
- Aggregate across coast: how many days under advisory/closure across all stations (sum of station totals); break down whether it's just a few troublesome sites (seems to be based on presentations I've seen from DEQ) and what % of advisory/closure days came from those few sites



About

The `.qmd` file that generated this section was: **beach_advisories.qmd**.

For information on data sources and processing, see *appendix cross-reference here*.

11 Shellfish Closures

This section will be based on data from MDMR.

11.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

e.g. "Shellfish beds were closed less than normal this year."

11.2 Supporting Information

11.2.1 DMR

About

The `.qmd` file that generated this section was: **shellfish_closures.qmd**.

For information on data sources and processing, see *appendix cross-reference here.*

12 Impaired Waterbodies

This section will be based on DEQ's 303d list.

12.1 Overall Summary

Note: These summaries are NOT factual. They are examples of how we might want to talk about what we find, to help ensure we do calculations in the right way. I am completely making up details and could be very, very wrong. Please do not treat this information as factual.

Here just the number of miles (rivers/streams) or acres (lakes, estuaries) that are listed as impaired. Was this more or less than the last State of the Sound report?

12.2 Supporting Information

12.2.1 DEQ Info

About

The `.qmd` file that generated this section was: **impaired_waterbodies.qmd**.

For information on data sources and processing, see *appendix cross-reference here.*

A Data Processing

In general, links to original data sources are provided in each section below. Because data files are routinely updated, the files as downloaded and used in this document are also available online. These files are provided for reproducibility and transparency; for your own analyses, we encourage you to seek out data from the original linked sources.

A.1 Weather and Climate-related variables

A.1.1 Precipitation

Go back to Chapter 1

A.1.2 Freshwater Inflows

Go back to Chapter 2

A.2 People and Land Use

A.2.1 Population

Go back to Chapter 3

A.2.2 Land Use / Land Cover

Go back to Chapter 4

A.3 Water Conditions

A.3.1 Salinity

Go back to Chapter 5

A.3.2 Dissolved Oxygen

Go back to Chapter 6

A.3.3 Turbidity / Suspended Solids

Go back to Chapter 7

A.3.4 Nutrients

Go back to Chapter 8

A.3.5 Chlorophyll *a*

Go back to Chapter 9

A.4 Pollutants/Contaminants

A.4.1 Beach Advisories

Go back to Chapter 10

A.4.2 Shellfish Closures

Go back to Chapter 11

A.4.3 Impaired Waterbodies

Go back to Chapter 12

About

The `.qmd` file that generated this section was: **data_processing.qmd**.